

Notes: Hou, Xue, and Zhang (RFS, 2015)

Digesting Anomalies: An Investment Approach

Contents

1	Big picture	2
2	Data and factor construction	2
2.1	Sample and returns	2
2.2	The investment-based equation	3
2.3	Factor definitions	3
2.4	Investment factor	3
2.5	Profitability factor	3
2.6	Anomaly universe	4
3	Empirical specifications and tests	4
3.1	High-minus-low anomaly portfolio	4
3.2	Factor regressions	4
3.3	Overall model performance	4
3.4	Sharpe-ratio comparison	5
4	Identification and interpretation	5
4.1	What the paper is trying to identify	5
4.2	Why investment and profitability are informative	5
4.3	Limits of the exercise	5
5	Tables and figures	6
5.1	Figure 1 and Table 1: the investment channel and factor properties	6
5.2	Table 2: the anomaly universe	6
5.3	Table 3 and Table 4: insignificant and significant anomalies	7
5.4	Table 5 and Table 6: the role of the size factor and factor loadings	8
5.5	Table 7 and Table 8: momentum and momentum reversal	9
5.6	Table 9: operating accruals and percent operating accruals	9
5.7	Table 10 and Table 11: size-B/M portfolios and Sharpe ratios	9
6	Short summary	10
7	Conclusion	11
7.1	Core conclusion	11
7.2	Economic intuition	11
7.3	Takeaway	11

1 Big picture

- **Question.** Can a small asset-pricing model, motivated by investment-based theory, digest the large zoo of return anomalies?
- **Main contribution.** The paper proposes and tests a four-factor q -factor model with market, size, investment, and profitability factors.
- **Economic idea.** In q -factor theory, expected returns are high when expected profitability is high and investment is low. Firms with high investment, all else equal, have low expected returns; firms with high profitability, all else equal, have high expected returns.
- **Factor set.** The factors are: market factor, size factor r_{ME} , investment factor $r_{I/A}$, and profitability factor r_{ROE} .
- **Testing universe.** The authors examine nearly 80 anomaly variables, grouped into momentum, value-versus-growth, investment, profitability, intangibles, and trading-friction categories.
- **Main empirical result.** Many anomalies that look significant under the Fama–French and Carhart models are absorbed by the q -factor model.
- **Broad cross-section result.** A large fraction of candidate anomalies are not significant in broad NYSE-breakpoint decile sorts. Among the significant ones, the q -factor model often produces smaller alphas and weaker GRS rejections than competing models.
- **Mechanism.** Many anomalies earn high-minus-low returns because their long-short portfolios load positively on profitability and negatively on investment.
- **Momentum result.** The model has difficulty fully digesting short-horizon momentum, especially earnings and price momentum, but performs better for longer-horizon reversals.
- **Accrual result.** The model struggles with operating accruals but performs better with percent operating accruals because the latter align more cleanly with profitability and investment exposures.
- **25 size-B/M portfolios.** The q -factor model also performs competitively on the classic 25 size and book-to-market portfolios.
- **Takeaway.** A large number of anomalies can be organized around investment and profitability, suggesting that many characteristic-based return patterns are manifestations of a small set of economic forces.

2 Data and factor construction

2.1 Sample and returns

- The empirical sample uses U.S. common stocks with CRSP and Compustat data.
- Monthly returns are used to form value-weighted testing portfolios.
- Most anomaly portfolios use NYSE breakpoints to avoid letting microcaps dominate the cross-section.
- The main sample for factor returns runs from January 1972 to December 2012, although some variables have shorter histories depending on data availability.

Interpretation: The NYSE-breakpoint design is important because many published anomalies are strongest among tiny, illiquid stocks. The authors ask whether anomalies matter in a broad investable cross-section.

2.2 The investment-based equation

The model's core intuition comes from the first-order condition for investment. In the paper's notation, the expected stock return satisfies

$$E_0[r_{i1}^S] = \frac{E_0[\Pi_{i1}]}{1 + a(I_{i0}/A_{i0})}. \quad (1)$$

- Π_{i1} is expected profitability.
- I_{i0}/A_{i0} is investment relative to assets.
- $a(\cdot)$ captures adjustment costs.

Interpretation: Expected returns rise with expected profitability and fall with investment. This equation is the bridge from q -factor theory to the empirical investment and ROE factors.

2.3 Factor definitions

The q -factor model includes four factors:

$$r_{i,t} - r_{f,t} = \alpha_i + \beta_{i,MKT}MKT_t + \beta_{i,ME}r_{ME,t} + \beta_{i,I/A}r_{I/A,t} + \beta_{i,ROE}r_{ROE,t} + \epsilon_{i,t}. \quad (2)$$

- MKT is the value-weighted market excess return.
- r_{ME} is a size factor: small-minus-big.
- $r_{I/A}$ is an investment factor: low investment-minus-high investment.
- r_{ROE} is a profitability factor: high ROE-minus-low ROE.

Interpretation: The signs are chosen so that high factor returns correspond to high expected-return characteristics: small size, low investment, and high profitability.

2.4 Investment factor

Investment-to-assets is computed as annual asset growth:

$$I/A = \frac{A_t - A_{t-1}}{A_{t-1}}. \quad (3)$$

Interpretation: Low investment firms should earn high average returns if low investment reflects high discount rates or limited positive-NPV opportunities.

2.5 Profitability factor

Profitability is measured with return on equity:

$$ROE = \frac{\text{income before extraordinary items}}{\text{one-quarter-lagged book equity}}. \quad (4)$$

Interpretation: High ROE firms should earn high returns if expected profitability is an important state variable in investment-based pricing.

2.6 Anomaly universe

The paper studies anomaly variables in six groups:

1. momentum;
2. value-versus-growth;
3. investment;
4. profitability;
5. intangibles;
6. trading frictions.

Interpretation: The breadth of the anomaly set lets the authors ask whether the q -factor model is a general organizing framework rather than a model designed for a few selected anomalies.

3 Empirical specifications and tests

3.1 High-minus-low anomaly portfolio

For each anomaly variable z , stocks are sorted into deciles using NYSE breakpoints. The main test asset is the high-minus-low portfolio:

$$r_{z,t}^{H-L} = r_{z,t}^{High} - r_{z,t}^{Low}. \quad (5)$$

Interpretation: If an anomaly is economically meaningful, the high-minus-low decile should earn a significant average return and an alpha that factor models cannot explain.

3.2 Factor regressions

For each decile or high-minus-low portfolio, the paper estimates alphas under several models:

$$r_{p,t} - r_{f,t} = \alpha_p^{CAPM} + \beta_p MKT_t + \epsilon_{p,t}, \quad (6)$$

$$r_{p,t} - r_{f,t} = \alpha_p^{FF} + \beta_p MKT_t + s_p SMB_t + h_p HML_t + \epsilon_{p,t}, \quad (7)$$

$$r_{p,t} - r_{f,t} = \alpha_p^C + \beta_p MKT_t + s_p SMB_t + h_p HML_t + m_p UMD_t + \epsilon_{p,t}, \quad (8)$$

$$r_{p,t} - r_{f,t} = \alpha_p^q + \beta_{p,MKT} MKT_t + \beta_{p,ME} r_{ME,t} + \beta_{p,I/A} r_{I/A,t} + \beta_{p,ROE} r_{ROE,t} + \epsilon_{p,t}. \quad (9)$$

Interpretation: The main question is whether α^q is small and insignificant even when traditional alphas remain large.

3.3 Overall model performance

The paper evaluates factor models with three summary diagnostics:

- average absolute alpha across deciles;
- Gibbons–Ross–Shanken test of whether all decile alphas are jointly zero;
- explanatory factor loadings on r_{ME} , $r_{I/A}$, and r_{ROE} .

Interpretation: A good model should not merely explain the high-minus-low spread; it should also price the full set of decile portfolios.

3.4 Sharpe-ratio comparison

For a vector of factor returns f , the maximum squared Sharpe ratio is

$$SR_{max}^2 = \mu_f' V_f^{-1} \mu_f, \quad (10)$$

where μ_f is the vector of factor means and V_f is the variance-covariance matrix.

Interpretation: The Sharpe-ratio exercise asks whether the new factors substantially enlarge the investment opportunity set compared with traditional factors.

4 Identification and interpretation

4.1 What the paper is trying to identify

- The paper is not a causal experiment.
- It is a model-evaluation and anomaly-digestion exercise.
- The identification question is whether a small number of economically motivated factors jointly explain a large number of anomaly returns.

Interpretation: The paper's strength is breadth: if many anomalies are absorbed by investment and profitability factors, then those factors are likely capturing common expected-return forces.

4.2 Why investment and profitability are informative

- Low investment and high profitability have direct interpretations in q -factor theory.
- They also subsume many characteristics that appear unrelated on the surface.
- For example, valuation, accruals, net issuance, and profitability anomalies often align with investment or ROE loadings.

Interpretation: The economic content is that expected returns are tied to firms' production and investment decisions, not only to covariance with consumption or market factors.

4.3 Limits of the exercise

- Factor construction is empirical and uses characteristics directly, so the model is not a pure structural test.
- Some anomalies, especially short-horizon momentum, remain difficult.
- Microcap sensitivity and data-mining concerns remain important for many anomalies.
- The model's success does not prove that all anomalies are risk-based; mispricing stories can also generate characteristic-based factors.

Interpretation: The paper narrows the anomaly zoo but does not settle the risk-versus-mispricing debate.

5 Tables and figures

The figures and tables below are directly cropped from the paper and grouped compactly.

5.1 Figure 1 and Table 1: the investment channel and factor properties

Read:

- Figure 1 summarizes the model’s core implication: high investment implies low expected returns, while high expected profitability implies high expected returns.
- Table 1 shows that r_{ME} earns 0.31% per month, $r_{I/A}$ earns 0.45%, and r_{ROE} earns 0.58%.
- The investment factor is highly correlated with HML, and the ROE factor is correlated with momentum.

Economic message: Investment and profitability are not arbitrary characteristics. They are theoretical state variables that also line up with familiar empirical factors.

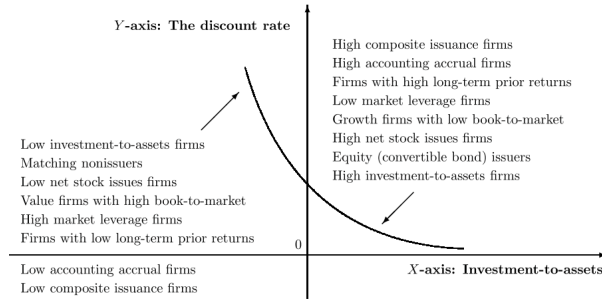


Figure 1
the investment channel

Figure 1: investment channel

Panel A: Descriptive statistics							Panel B: Correlation matrix (q-values)							
	Mean	σ	β_{MKT}	β_{SMB}	β_{HML}	β_{UMD}	R^2		$r_{I/A}$	r_{ROE}	MKT	SMB	HML	UMD
r_{ME} (2.12)	0.31 (1.62)	0.23 (1.43)	0.17 (4.33)	0.02 (0.04)	0.09 (57.37)	0.17 (7.05)	0.06 (0.02)	r_{ME}	-0.11 (0.02)	-0.31 (0.00)	0.25 (0.00)	0.95 (0.13)	0.95 (0.90)	0.01 (0.91)
	0.40 (1.59)	0.42 (5.73)	0.09 (7.77)	0.00 (0.00)	0.17 (57.37)	0.17 (7.05)	0.93 (0.20)	$r_{I/A}$	0.06 (0.20)	-0.26 (0.00)	-0.22 (0.00)	-0.09 (0.00)	-0.09 (0.31)	
	0.01 (0.15)	0.02 (2.40)	0.01 (0.51)	0.01 (0.51)	0.19 (65.51)	0.03 (7.34)	0.03 (2.16)	r_{ROE}	-0.19 (0.00)	-0.36 (0.00)	-0.36 (0.00)	-0.36 (0.00)	-0.36 (0.00)	
	0.45 (4.95)	0.52 (5.03)	-0.06 (-5.58)	-0.02 (-0.81)	0.39 (11.98)	0.03 (1.41)	0.93 (0.20)	MKT	0.28 (0.02)	0.28 (0.02)	0.32 (0.00)	-0.23 (-0.15)	-0.15 (0.79)	
$r_{I/A}$	0.45 (4.95)	0.52 (5.03)	-0.06 (-5.58)	-0.02 (-0.81)	0.39 (11.98)	0.03 (1.41)	0.93 (0.20)	SMB				-0.23 (-0.15)	-0.15 (0.79)	
	0.33 (0.85)	0.36 (3.06)	-0.02 (-0.91)	0.39 (11.98)	0.03 (1.41)	0.93 (0.20)	0.50 (0.05)	HML					-0.09 (0.00)	
	0.28 (3.58)	0.35 (3.24)	-0.02 (-0.87)	0.41 (11.94)	0.05 (1.97)	0.05 (1.97)	0.04 (0.04)						0.00 (0.00)	
	0.58 (4.81)	0.63 (5.62)	-0.11 (-2.38)	-0.05 (-0.87)	0.41 (11.94)	0.05 (1.97)	0.04 (0.04)						0.00 (0.00)	
r_{ROE}	0.58 (4.81)	0.63 (5.62)	-0.11 (-2.38)	-0.05 (-0.87)	0.41 (11.94)	0.05 (1.97)	0.04 (0.04)						0.00 (0.00)	
	0.77 (6.84)	0.77 (6.84)	-0.09 (-2.08)	-0.33 (-5.75)	-0.20 (-2.38)	0.20 (0.20)	0.20 (0.20)						0.20 (0.20)	
	0.50 (4.75)	0.50 (4.75)	-0.03 (-1.38)	-0.33 (-5.75)	-0.10 (-1.48)	0.28 (6.73)	0.28 (6.73)						0.28 (6.73)	
	0.50 (4.75)	0.50 (4.75)	-0.03 (-1.38)	-0.33 (-5.75)	-0.10 (-1.48)	0.28 (6.73)	0.28 (6.73)						0.28 (6.73)	

Size (ME) is price per share times shares outstanding. Investment-to-assets (I/A) is the annual change in total assets (Compustat annual item AT) divided by lagged total assets. ROE is income before extraordinary items (Compustat quarterly item IBQ) divided by 1-quarter-lagged book equity. Book equity is the shareholders' equity, plus balance-sheet deferred taxes and investment tax credits (item TADTCQ) if available, minus the book value of preferred stock. Depending on availability, we use the stockholders' equity (item SEQ), common equity (item CEQ), or total assets (item ATQ) minus liabilities (item LTQ) in that order as the shareholders' equity. We use redemption value (item PSTKRQ) if available, or carrying value for the book value of the preferred stock. At the end of June of each year t , we use the median NYSE size at the end of June to split NYSE, Amex, and NASDAQ stocks into 2 groups, small and big. Independently, at the end of June of each year t , we also sort stocks into 3 I/A groups, using the NYSE breakpoints for the low 30%, middle 40%, and high 30% of the ranked I/A for the fiscal year ending in calendar year $t-1$. In addition, independently, the beginning of each month, we sort stocks into 3 groups based on NYSE breakpoints for the low 30%, middle 40%, and high 30% of the ranked ROE. Earnings data in Compustat quarterly files are used in the sorts in the months immediately after the most recent public quarterly earnings announcement dates (item RDQ). Taking the intersections of the size, I/A , and 3 ROE groups, we form 18 portfolios. Monthly value-weighted returns on the 18 portfolios are calculated for the current month, and the portfolios are rebalanced monthly. The size factor, $size_t$, is the difference (small-minus-big), each month, between the average returns on the 9 small portfolios and the average returns on the 9 big portfolios.

Table 1: empirical properties of the q-factor factors

5.2 Table 2: the anomaly universe

Read:

- Table 2 lists the primary anomaly variables tested in the paper.
- The variables cover momentum, value-versus-growth, investment, profitability, intangibles, and trading frictions.
- The breadth of the table is important: the authors are not testing a narrow selected set of anomalies.

Economic message: The q -factor model is evaluated against a broad anomaly zoo, not a small hand-picked set of tests.

List of anomalies

Panel A: Momentum			
SUE-1	Earnings surprise (1-month holding period), Foster, Olsen, and Shevlin (1984)	SUE-6	Earnings surprise (6-month holding period), Foster, Olsen, and Shevlin (1984)
Abr-1	Cumulative abnormal stock returns around earnings announcements (1-month holding period), Chan, Jegadeesh, and Lakonishok (1996)	Abr-6	Cumulative abnormal stock returns around earnings announcements (6-month holding period), Chan, Jegadeesh, and Lakonishok (1996)
RE-1	Revisions in analysts' earnings forecasts (1-month holding period), Chan, Jegadeesh, and Lakonishok (1996)	RE-6	Revisions in analysts' earnings forecasts (6-month holding period), Chan, Jegadeesh, and Lakonishok (1996)
R6-1	Price momentum (6-month prior returns, 1-month holding period), Jegadeesh and Titman (1993)	R6-6	Price momentum (6-month prior returns, 6-month holding period), Jegadeesh and Titman (1993)
R11-1	Price momentum (11-month prior returns, 1-month holding period), Fama and French (1996)	I-Mom	Industry momentum, Moskowitz and Grinblatt (1999)
Panel B: Value-versus-growth			
B/M	Book-to-market equity, Rosenberg, Reid, and Lanstein (1985)	A/ME	Market leverage, Bhandari (1988)
Rev	Reversal, De Bondt and Thaler (1985)	E/P	Earnings-to-price, Basu (1983)
EF/P	Analysts' earnings forecasts-to-price, Elgers, Lo, and Pfeiffer (2001)	CF/P	Cash flow-to-price, Lakonishok, Shleifer, and Vishny (1994)
D/P	Dividend yield, Litzenger and Ramaswamy (1979)	O/P	Payout yield, Boudoukh et al. (2007)
NO/P	Net payout yield, Boudoukh et al. (2007)	SG	Sales growth, Lakonishok, Shleifer, and Vishny (1994)
LTG	Long-term growth forecasts of analysts, La Porta (1996)	Dur	Equity duration, Dechow, Sloan, and Soliman (2004)
Panel C: Investment			
ACI	Abnormal corporate investment, Titman, Wei, and Xie (2004)	I/A	Investment-to-assets, Cooper, Gulen, and Schill (2008)
NOA	Net operating assets, Hirshleifer et al. (2004)	$\Delta PI/A$	Changes in property, plant, and equipment plus changes in inventory scaled by assets, Lyandres, Sun, and Zhang (2008)
IG	Investment growth, Xing (2008)	NSI	Net stock issues, Pontiff and Woodgate (2008)
CEI	Composite issuance, Daniel and Titman (2006)	NXF	Net external financing, Bradshaw, Richardson, and Sloan (2006)
IvG	Inventory growth, Belo and Lin (2011)	IvC	Inventory changes, Thomas and Zhang (2002)
OA	Operating accruals, Sloan (1996)	TA	Total accruals, Richardson et al. (2005)
POA	Percent operating accruals, Hafzalla, Lundholm, and Van Winkle (2011)	PTA	Percent total accruals, Hafzalla, Lundholm, and Van Winkle (2011)
Panel D: Profitability			
ROE	Return on equity, Haugen and Baker (1996)	ROA	Return on assets, Balakrishnan, Bartov, and Fama (2010)
RNA	Return on net operating assets, Soliman (2008)	PM	Profit margin, Soliman (2008)
ATO	Asset turnover, Soliman (2008)	CTO	Capital turnover, Haugen and Baker (1996)
GP/A	Gross profits-to-assets, Novy-Marx (2013)	F	F -score, Piotroski (2000)
TES	Tax expense surprise, Thomas and Zhang (2011)	TI/BI	Taxable income-to-book income, Green, Hand, and Zhang (2013)
RS	Revenue surprise, Jegadeesh and Livnat (2006)	NEI	Number of consecutive quarters with earnings increases, Barth, Elliott, and Finn (1999)
FP	Failure probability, Campbell, Hilscher, and Szilagyi (2008)	O	O -score, Dichev (1998)

Table 2, panels A–D

Continued

Panel E: Intangibles			
OC/A	Organizational capital-to-assets, Eisfeldt and Papanikolaou (2013)	BC/A	Brand capital-to-assets, Belo, Lin, and Vitorino (2014)
Ad/M	Advertisement expense-to-market, Chan, Lakonishok, and Sougiannis (2001)	RD/S	R&D-to-sales, Chan, Lakonishok, and Sougiannis (2001)
RD/M	R&D-to-market, Chan, Lakonishok, and Sougiannis (2001)	RC/A	R&D capital-to-assets, Li (2011)
H/N	Hiring rate, Belo, Lin, and Bazdresch (2014)	OL	Operating leverage, Novy-Marx (2011)
G	Corporate governance, Gompers, Ishii, and Metrick (2003)	AccQ	Accrual quality, Francis et al. (2005)
Ind	Industries, Fama and French (1997)		
Panel F: Trading frictions			
ME	The market equity, Banz (1981)	Ivol	Idiosyncratic volatility, Ang et al. (2006)
Tvol	Total volatility, Ang et al. (2006)	Svol	Systematic volatility, Ang et al. (2006)
MDR	Maximum daily return, Bali, Cakici, and Whitelaw (2011)	β	Market beta, Frazzini and Pedersen (2014)
D- β	Dimson's beta, Dimson (1979)	S-Rev	Short-term reversal, Jegadeesh (1990)
Disp	Dispersion of analysts' earnings forecasts, Diether, Malloy, and Scherbina (2002)	Turn	Share turnover, Datar, Naik, and Radcliffe (1998)
I/P	1/share price, Miller and Scholes (1982)	Dvol	Dollar trading volume, Brennan, Chordia, and Subrahmanyam (1998)
Illiq	Illiquidity as absolute return-to-volume, Amihud (2002)		

Table 2, panels E–F

5.3 Table 3 and Table 4: insignificant and significant anomalies

Read:

- Table 3 shows that many prominent anomalies are insignificant in broad NYSE-breakpoint value-weighted deciles.
- Table 4 reports model pricing performance for 35 significant anomalies.
- The q -factor model generally produces smaller alphas than the CAPM, Fama–French, and Carhart models.
- Only a subset of significant anomaly spreads retain significant q -factor alphas.

Economic message: Many anomalies either disappear in a broad investable cross-section or are substantially absorbed by the q -factor model.

Significant anomalies in the broad cross section

	R6-1	A/ME	Rev	EF/P	D/P	O/P	SG	LTG	ACI	NXF
m	0.48	0.43	-0.39	0.45	0.27	0.35	-0.27	0.01	-0.27	-0.30
t_m	1.43	1.82	-1.57	1.73	0.94	1.53	-1.34	0.02	-1.70	-1.55
	TA	RNA	PM	ATO	CTO	F	TES	TI/BI	RS	O
m	-0.19	0.13	0.10	0.22	0.20	0.37	0.32	0.13	0.29	-0.08
t_m	-1.31	0.61	0.40	1.11	1.11	1.28	1.92	0.86	1.82	-0.37
	BC/A	RD/S	RC/A	H/N	G	AccQ	ME	Ivol	Tvol	MDR
m	0.18	0.01	0.32	-0.25	0.03	-0.18	-0.24	-0.54	-0.37	-0.31
t_m	0.73	0.06	1.27	-1.47	0.09	-0.79	-0.90	-1.56	-0.95	-0.94
	β	D- β	S-Rev	Disp	Turn	I/P	Dvol	Illiq		
m	-0.13	0.07	-0.31	-0.33	-0.12	-0.00	-0.26	0.27		
t_m	-0.36	0.30	-1.39	-1.24	-0.43	-0.01	-1.30	1.14		

For each anomaly variable, we report the average return (m) of the high-minus-low decile and its t -statistic (t_m) adjusted for heteroscedasticity and autocorrelations. Table 2 provides a brief description of the symbols. We form all of the deciles using NYSE breakpoints, and calculate value-weighted portfolio returns. Appendix A details variable definition and portfolio construction.

Table 4
Significant anomalies in the broad cross section: Pricing errors and tests of overall performance

	SUE-1	SUE-6	Abt-1	Abt-6	RE-1	RE-6	R1-1	I-Mom	B/M	EP	CFP	NOVP	Dur	I/A	NOA	$\Delta P/A$	IG
α	0.45	0.24	0.71	0.30	0.09	0.60	0.85	1.18	0.51	0.70	1.59	0.52	0.66	-0.54	-0.42	-0.38	-0.41
σ	0.50	0.27	0.76	0.31	1.02	0.71	0.92	1.29	0.58	0.75	0.69	0.63	0.84	-0.62	-0.50	-0.38	-0.57
σ_{FF}	0.55	0.39	0.84	0.38	1.20	0.84	1.12	1.52	0.68	0.01	0.05	0.01	0.52	-0.06	-0.15	-0.52	-0.41
σ_C	0.34	0.18	0.62	0.19	0.56	0.37	0.06	0.09	-0.18	-0.01	0.01	0.06	0.49	0.08	0.09	0.41	0.36
σ_q	0.16	0.02	0.64	0.26	0.12	0.03	0.24	0.00	0.21	0.17	0.22	0.36	-0.27	0.14	-0.38	-0.26	0.05
σ_n	5.99	2.17	5.50	5.11	5.43	2.98	5.17	5.52	2.53	2.88	2.63	2.44	3.23	-2.90	-2.45	-2.55	-1.43
t	4.26	2.68	5.84	3.33	4.13	3.28	3.63	4.18	2.68	3.05	3.12	3.01	4.45	-2.98	-2.94	-2.52	-1.91
t_{FF}	4.50	3.62	5.93	3.99	4.81	4.52	4.47	4.99	3.25	0.34	0.08	0.51	2.64	0.09	0.55	0.59	0.99
t_C	2.62	1.69	4.37	2.06	2.56	2.15	0.51	0.67	-1.11	-0.06	0.03	-0.40	3.33	-0.56	-0.61	-2.69	-1.51
t_q	1.12	0.18	4.07	2.18	0.43	0.14	0.71	0.54	0.01	1.15	0.76	1.04	2.38	-1.32	1.08	-1.90	-1.85
t_n	0.16	0.11	0.13	0.08	0.19	0.14	0.17	0.21	0.16	0.22	0.23	0.20	0.23	0.24	0.17	0.15	0.13
t_{FF}^*	0.17	0.13	0.16	0.11	0.27	0.23	0.19	0.26	0.15	0.07	0.10	0.08	0.17	0.11	0.12	0.17	0.13
t_C^*	0.11	0.09	0.12	0.08	0.11	0.09	0.10	0.13	0.06	0.09	0.07	0.15	0.08	0.10	0.14	0.12	0.11
t_q^*	0.05	0.07	0.13	0.07	0.10	0.11	0.08	0.13	0.13	0.08	0.10	0.14	0.12	0.08	0.09	0.12	0.14
t_n^*	0.00	0.00	0.00	0.01	0.04	0.21	0.00	0.09	0.04	0.01	0.05	0.00	0.00	0.00	0.00	0.00	0.00
t_{FF}^*	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.19	0.18	0.43	0.00	-0.15	0.01	0.00	0.00
t_C^*	0.00	0.00	0.00	0.01	0.16	0.12	0.00	0.00	0.45	0.29	0.38	0.37	0.00	0.41	0.02	0.00	0.01
t_q^*	0.42	0.04	0.00	0.02	0.46	0.08	0.00	0.01	0.03	0.15	0.13	0.02	0.00	0.72	0.01	0.00	0.01

(continued)

Table 4
Continued

	NSI	CEI	IvG	IvC	OA	POA	PTA	ROE	ROA	GPA	NEI	FP	OC/A	AdM	RDM	OL	Svol
α	-0.68	-0.57	-0.41	-0.45	-0.30	-0.46	-0.40	0.80	0.62	0.34	0.39	-0.67	0.56	0.79	0.63	0.39	-0.60
σ	-0.78	-0.79	-0.47	-0.51	-0.33	-0.53	-0.50	0.96	0.78	0.32	0.40	-1.06	0.65	0.82	0.47	0.44	-0.72
σ_{FF}	-0.64	-0.50	-0.29	-0.38	-0.37	-0.32	-0.29	1.17	1.00	0.50	0.63	-1.44	0.61	0.15	0.22	0.37	-0.66
σ_C	-0.54	-0.40	-0.19	-0.30	-0.31	-0.25	-0.27	0.85	0.67	0.45	0.43	-0.67	0.40	0.32	0.31	0.33	-0.62
σ_q	-0.26	-0.22	-0.03	-0.28	-0.56	-0.12	-0.10	0.05	0.09	0.11	0.18	-0.17	0.09	0.11	0.60	-0.05	-0.37
σ_n	-4.13	-3.96	-2.77	-3.05	-2.32	-3.02	-2.57	3.11	2.70	2.18	3.31	-1.08	4.07	2.96	2.31	2.06	-2.57
t	-4.86	-4.79	-3.29	-3.35	-2.47	-3.64	-3.50	4.02	3.67	2.02	3.45	-3.80	4.69	3.08	1.81	2.22	-3.12
t_{FF}	-4.28	-3.72	-2.10	-2.61	-2.84	-2.42	-2.06	5.43	5.40	3.25	6.03	-6.44	4.52	0.79	0.03	1.91	-2.80
t_C	-3.58	-3.93	-1.34	-1.97	-2.32	-1.88	-1.82	4.03	3.59	2.85	3.73	-3.79	2.97	1.37	1.40	1.76	-2.59
t_q	-1.75	-1.80	-0.20	-1.84	-2.00	-0.87	-0.67	0.37	0.71	1.68	-0.57	0.66	0.39	2.40	-0.27	-1.42	0.48
t_n	0.18	0.19	0.14	0.16	0.15	0.12	0.12	0.18	0.15	0.06	0.19	0.16	0.14	0.23	0.13	0.11	0.18
t_{FF}^*	0.18	0.15	0.11	0.12	0.13	0.11	0.11	0.24	0.23	0.14	0.23	0.23	0.15	0.13	0.17	0.11	0.19
t_C^*	0.15	0.15	0.10	0.10	0.12	0.11	0.10	0.15	0.14	0.14	0.15	0.12	0.13	0.18	0.21	0.12	0.16
t_q^*	0.11	0.12	0.11	0.08	0.15	0.12	0.08	0.09	0.07	0.11	0.09	0.13	0.11	0.11	0.27	0.12	0.11
t_n^*	0.00	0.00	0.01	0.00	0.00	0.01	0.01	0.07	0.25	0.00	0.04	0.24	0.54	0.04	0.04	0.04	0.01
t_{FF}^*	0.00	0.00	0.01	0.00	0.00	0.01	0.00	0.00	0.00	0.01	0.00	0.00	0.00	0.00	0.18	0.02	0.07
t_C^*	0.00	0.00	0.11	0.04	0.00	0.01	0.02	0.00	0.04	0.01	0.00	0.00	0.00	0.07	0.01	0.06	0.06
t_q^*	0.02	0.01	0.08	0.56	0.00	0.00	0.11	0.05	0.00	0.01	0.05	0.00	0.02	0.07	0.00	0.09	0.30

Table 4, part 1

Table 4, part 2

5.4 Table 5 and Table 6: the role of the size factor and factor loadings

Read:

- Table 5 evaluates an alternative q -factor model without the size factor.
- Table 6 shows how significant high-minus-low anomaly portfolios load on market, size, investment, and ROE factors.
- Many anomalies have economically meaningful loadings on investment and profitability.
- The characteristics of the long-short portfolios often line up with these factor exposures.

Economic message: The factor loadings explain why the model works: anomaly returns are often compensation for exposure to low investment and high profitability.

Table 5
Significant anomalies in the broad cross section: An alternative q -factor model without the size factor

	SUE-1	SUE-6	Abt-1	Abt-6	RE-1	RE-6	R6-6	R1-1	I-Mom	B/M	EP	CFP	NOVP	Dur	I/A	NOA	$\Delta P/A$	IG
α_1^H	0.21	0.07	0.67	0.30	0.05	-0.05	0.37	0.42	0.15	0.42	0.29	0.30	0.21	-0.38	0.09	-0.35	-0.28	0.00
α_2^H	1.43	0.42	3.95	2.48	0.18	-0.22	0.96	0.85	0.48	2.13	1.21	1.42	1.26	-1.72	0.68	-1.71	-2.00	0.00
α_3^H	0.05	0.07	0.15	0.08	0.14	0.15	0.10	0.16	0.09	0.11	0.12	0.15	0.12	0.11	0.09	0.12	0.11	0.13
α_4^H	0.62	0.13	0.90	0.00	0.20	0.02	0.00	0.00	0.10	0.15	0.04	0.01	0.01	0.31	0.01	0.01	0.05	0.00
NSI																		
CEI																		
IvG																		
IvC																		
OA																		
POA																		
PTA																		
ROE																		
ROA																		
GPA																		
NEI																		
FP																		
OC/A																		
AdM																		
RDM																		
OL																		
Svol																		

For each anomaly variable, α_i^H is the high-minus-low alpha from an alternative 3-factor q -model without the size factor, and t_i^H is the t -statistic adjusted for heteroscedasticity and autocorrelations. $|t_i^H|$ is the maximum absolute value of the alphas across a return set of deciles, and α_i^H is the α -value of the GRS test on the null hypothesis that all of the alphas are jointly zero.

Table 6
Factor loadings and underlying characteristics for significant anomalies in the broad cross section

	SUE-1	SUE-6	Abt-1	Abt-6	RE-1	RE-6	R6-6	R1-1	I-Mom	B/M	EP	CFP	NOVP	Dur	I/A	NOA	$\Delta P/A$	IG	
β_{PACT}	-0.08	-0.06	-0.06	-0.03	-0.05	-0.07	-0.09	-0.14	-0.11	-0.03	-0.12	-0.15	-0.18	0.11	0.02	-0.02	0.05	-0.02	
β_{PAC}	0.10	0.09	0.07	0.09	-0.15	-0.19	0.27	0.40	0.31	0.46	0.25	0.19	-0.32	-0.23	-0.11	0.06	-0.05	-0.11	
$\beta_{P/A}$	0.02	-0.11	-0.13	-0.16	0.04	-0.12	-0.07	0.04	-0.03	1.45	0.99	1.01	1.03	-0.85	-1.37	-0.01	-0.77	-0.82	
$\beta_{P/ROE}$	0.48	0.45	0.28	0.18	1.33	1.12	1.02	1.48	0.82	-0.51	-0.09	-0.24	0.02	0.24	0.15	-0.01	0.16	-0.07	
$\beta_{P/ROA}$	-1.82	-1.53	-1.31	-1.20	-0.76	-1.24	-1.17	-1.43	-1.72	-0.59	-2.02	-2.41	-3.86	1.67	0.62	-0.55	1.35	-0.71	
$\beta_{P/SG}$	1.94	1.27	0.67	1.82	-1.42	-1.98	1.43	1.74	1.86	5.37	1.90	1.66	-4.40	-1.61	-1.81	0.54	-0.94	-1.95	
$\beta_{P/IG}$	0.18	-0.97	-1.25	-2.24	0.25	-0.82	-0.27	0.12	-0.13	12.74	5.76	6.79	10.25	-5.69	-15.50	0.			
$\beta_{P/IG}^*$	5.75	5.95	1.25	2.24	204	100.9	9.96	5.31	5.67	4.00	-5.58	-0.66	-0.17	0.19	1.87	0.54	-0.12	1.93	-1.06
$\beta_{P/ROE}^*$	0.69	0.73	-0.01	0.03	0.77	0.87	0.74	0.54	0.52	0.62	-2.74	0.73	0.28	1.24	0.88	0.07	0.63	0.23	
$\beta_{P/ROA}^*$	-1.46	-1.06	1.37	1.13	-0.80	-0.87	-0.47	-0.83	-1.18	-0.70	-1.11	-5.63	-14.43	3.99	-5.65	0.16	-5.61	-6.10	
$\beta_{P/SG}^*$	1.24	0.97	-0.01	-0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	
$\beta_{P/IG}^*$	0.91	4.38	-0.29	1.31	8.75	96.53	49.2	4.95	3.67	-10.31	4.09	-4.57	7.74	5.23	7.75	1.71	7.56	6.16	
$\beta_{P/IG}^{**}$	1.46	1.07	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	
$\beta_{P/IG}^{***}$	16.46	10.97	13.38	15.47	29.97	27.86	1.60	17.06	2.04	-29.57	1.30	4.83	7.27	2.22	10.00	-7.89	5.10	3.23	
	NSI	CEI	IG	LC	CA	POA	PAA	ROA	ROE	ROA	GPA	NIA	FP	OCIA	ADM	RDIM	OL	Sveol	
β_{PACT}	0.04	0.24	-0.03	0.04	0.03	0.01	0.06	-0.10	-0.14	0.05	-0.02	0.02	0.04	0.13	0.04	0.16	-0.06	0.04	
β_{PAC}	0.17	0.26	0.01	0.01	-0.21	0.28	0.21	0.28	0.28	0.28	0.28	0.28	0.28	0.28	0.28	0.28	0.28	0.28	
$\beta_{P/A}$	-0.08	-0.16	-0.06	-0.06	0.02	0.02	0.90	0.90	-0.10	-0.14	0.24	-0.30	0.17	0.35	1.42	0.21	0.16	-0.21	
$\beta_{P/ROE}$	-0.32	-0.12	0.05	0.18	0.29	0.05	0.10	0.10	0.13	0.62	0.63	1.61	0.51	0.27	0.58	0.54	0.43	-0.03	
$\beta_{P/ROA}$	0.99	0.27	0.01	0.01	0.01	0.01	0.90	1.50	1.50	1.50	1.50	1.50	1.50	1.50	1.50	1.50	1.50	1.50	
$\beta_{P/SG}$	-2.24	3.79	2.85	-1.07	4.41	3.28	-6.56	-4.61	6.51	2.53	2.45	2.49	5.96	5.96	2.85	6.75	2.63	2.30	
$\beta_{P/IG}$	0.18	-1.42	0.56	-1.99	6.20	6.21	0.04	0.05	0.27	16.66	7.00	5.35	8.79	7.12	-1.37	-4.10	-4.85	-3.54	
$\beta_{P/IG}^*$	-1.37	-1.79	-0.26	0.24	-0.24	-0.36	-0.36	-2.81	2.66	6.39	2.34	3.00	-1.31	-1.34	-4.39	-1.31	-0.19	-0.19	
$\beta_{P/ROE}^*$	-0.01	-0.01	-0.01	-0.01	-0.01	-0.01	-0.01	-0.01	-0.01	-0.01	-0.01	-0.01	-0.01	-0.01	-0.01	-0.01	-0.01	-0.01	
$\beta_{P/ROA}^*$	-1.71	-1.41	0.42	0.07	0.88	0.02	0.36	16.95	14.71	3.94	4.32	7.84	1.52	-3.33	-2.80	1.86	-0.64	-0.64	
$\beta_{P/SG}^*$	-0.64	-7.77	4.27	4.28	-6.48	-5.13	-3.99	-6.76	11.56	0.50	0.88	14.58	-0.76	-9.44	-9.83	-9.47	-8.43	-9.43	
$\beta_{P/IG}^{**}$	-0.85	-1.37	0.42	0.07	0.88	0.02	0.36	16.95	14.71	3.94	4.32	7.84	1.52	-3.33	-2.80	1.86	-0.64	-0.64	
$\beta_{P/IG}^{***}$	-11.87	-0.07	3.31	8.13	5.16	7.42	2.56	20.92	27.97	3.88	3.76	-25.96	7.79	-12.60	-9.21	11.42	-4.01	-4.01	

5.5 Table 7 and Table 8: momentum and momentum reversal

Read:

- Table 7 shows that earnings momentum and price momentum remain difficult for the q -factor model.
- Momentum deciles still show strong return patterns and sizable alphas.
- Table 8 studies longer-horizon reversal portfolios and shows that short-horizon momentum effects fade over time.

Economic message: Momentum is the most important limitation of the q -factor model. The model performs better when momentum has begun to reverse.

	Panel A: Earnings momentum										Panel B: Price momentum				
	SUE-12					SUE-13-36					R6-12		R6-13-36		R6-37-60
	Low	2	3	4	5	6	7	8	9	High					
m	0.36	0.54	0.55	0.28	0.44	0.43	0.64	0.64	0.64	0.80	0.16	0.13	0.57	-0.24	-0.03
a	-0.13	-0.15	-0.15	-0.20	-0.01	-0.03	0.29	0.18	0.19	0.38	0.19	-0.08	0.16	-0.25	-0.05
α_{FF}	-0.12	-0.15	-0.15	-0.18	-0.01	-0.02	0.21	0.24	0.21	0.43	0.31	-0.04	0.14	-0.04	0.07
α_C	0.00	-0.06	-0.07	-0.10	0.01	0.02	0.21	0.16	0.15	0.34	-0.03	0.16	0.09	-0.02	0.06
α_q	1.47	1.41	1.36	1.17	2.00	1.86	3.01	3.06	2.96	3.89	0.06	1.02	1.89	2.38	2.24
t_m	-1.39	-1.62	-1.86	-2.30	-0.18	-0.15	2.58	2.54	2.64	5.14	-3.60	-2.07	-0.28	1.00	0.47
t_{FF}	-1.29	-1.73	-1.75	-2.30	-0.08	-0.28	2.81	3.43	2.87	5.86	-4.34	-2.49	-0.89	0.40	-0.23
t_C	0.01	-0.09	-0.80	-1.18	0.07	0.29	2.61	2.13	1.69	4.55	-0.29	2.18	4.24	4.27	1.77
The q -factor model regressions										The q -factor model regressions					
α_q	0.05	0.00	0.04	0.05	0.00	-0.03	0.09	0.02	0.04	0.21	0.00	0.04	0.11	0.11	-0.01
β_{MKT}	1.03	1.00	1.02	0.94	0.96	0.98	1.01	0.97	0.95	1.19	1.05	1.00	0.96	0.94	0.95
β_{RM}	-0.16	0.04	0.00	-0.02	0.00	-0.04	-0.05	-0.03	0.01	-0.05	0.16	-0.04	-0.08	-0.06	-0.05
$\beta_{I/A}$	0.00	-0.16	-0.12	-0.25	0.06	0.03	0.06	0.07	0.06	0.02	-0.34	-0.08	0.02	0.07	0.09
β_{ROE}	-0.22	-0.12	-0.20	-0.19	-0.06	-0.01	0.11	0.21	0.20	0.28	-0.74	-0.36	-0.20	-0.09	0.00
t_q	0.42	-0.04	0.37	0.55	-0.04	-0.32	1.21	0.31	0.44	2.63	-0.02	0.24	0.92	1.21	-0.09
t_{RM}	32.75	39.64	39.24	36.70	43.64	41.42	51.90	55.30	42.68	37.45	24.15	25.44	35.15	40.62	42.64
$t_{I/A}$	-3.61	0.97	0.09	0.55	0.09	-0.85	-1.37	-1.22	0.18	-1.50	1.42	-0.48	-1.23	-1.76	-1.22
t_{ROE}	-0.01	-2.39	-1.99	-3.50	1.04	0.48	1.12	1.23	0.99	0.36	-2.39	-0.69	0.21	0.95	1.65
β_{ROE}	-3.19	-2.56	-3.62	-3.48	-1.53	-0.20	2.81	5.77	4.15	7.30	-5.87	-3.93	-2.86	-1.51	0.05
Characteristics in the q -factor model										Characteristics in the q -factor model					
ME	1.51	1.53	1.38	1.41	1.39	1.64	1.93	1.85	1.77	2.20	0.48	1.15	1.61	1.90	2.07
I/A	11.60	11.42	10.26	8.54	6.59	7.13	7.96	8.26	9.00	10.14	12.62	10.46	9.48	8.53	8.79
ROE	-0.80	1.94	2.05	2.34	2.72	3.24	3.59	3.75	3.95	5.00	-0.71	1.73	2.45	2.74	2.94

m , α_{FF} , α_C , and α_q are the average excess returns, the CAPM alpha, the Fama-French alpha, the Carhart alpha, and the q -alpha, and t_m , t_{FF} , t_C , and t_q are their t -statistics, respectively. β_{MKT} , β_{RM} , $\beta_{I/A}$, and β_{ROE} are the loadings on the market, size, investment, and ROE factors in the q -factor model, and β_{MKT} , β_{RM} , $\beta_{I/A}$, and β_{ROE} are their t -statistics, respectively. ME is the average market capitalization (in billions of dollars) across all of the firms in a given decile. I/A (in percent) is the sum of changes in assets across all of the firms in a given decile divided by the sum of their 1-year-lagged assets. ROE (in percent) is the sum of the most recently announced quarterly earnings across all of the firms in a given decile divided by the sum of their 1-year-lagged assets. All t -statistics are adjusted for heteroskedasticity and autocorrelations.

5.6 Table 9: operating accruals and percent operating accruals

Read:

- Table 9 compares the operating accrual anomaly with the percent operating accrual anomaly.
- The q -factor model struggles with operating accruals because the factor loadings go in the wrong direction.
- It performs better for percent operating accruals because the ROE and investment exposures align more naturally with the spread.

Economic message: The model's success depends on whether the anomaly portfolio's economic exposures correspond to the investment and profitability mechanisms.

Operating accrual (OA) and percent operating accrual (POA) data																															
		Panel A: OA										Panel B: POA																			
		Low					2					3					4					5					High				
m		0.50	0.56	0.66	0.60	0.57	0.60	0.52	0.40	0.40	0.20	0.64	0.56	0.60	0.61	0.46	0.52	0.55	0.43	0.39	0.18										
a		-0.06	0.08	0.22	0.18	0.16	0.20	0.07	-0.03	-0.08	-0.39	0.12	0.07	0.16	0.14	0.00	0.09	0.09	-0.03	-0.10	-0.41										
α_{FF}		0.10	0.14	0.22	0.13	0.11	0.16	0.10	-0.04	-0.05	-0.37	-0.01	0.03	0.13	0.12	0.01	0.11	0.22	0.08	-0.06	-0.32										
α_C		0.11	0.19	0.20	0.08	0.13	0.13	0.10	0.00	-0.02	-0.22	-0.03	0.05	0.07	0.15	0.00	0.12	0.22	0.12	-0.02	-0.28										
t_m		1.71	2.38	3.15	2.97	2.77	3.07	2.40	1.76	1.67	0.65	2.39	2.35	2.82	2.60	2.04	2.49	2.53	1.87	1.62	0.58										
t_a		-0.50	0.81	2.06	2.38	2.01	2.51	1.80	-0.30	-0.96	-3.65	1.04	0.77	1.92	1.75	0.05	1.32	1.27	-0.28	-1.09	-3.26										
t_{FF}		0.86	1.47	3.11	1.77	1.40	2.05	1.53	-0.47	-0.65	-3.22	-0.05	0.33	1.48	1.40	0.19	1.53	3.13	0.86	-0.73	-3.32										
t_C		0.98	1.74	2.58	1.08	1.54	1.56	1.46	0.02	-0.21	-2.49	-0.30	0.50	0.75	1.68	0.06	1.85	3.20	1.15	-0.22	-2.95										
The q -factor model regressions																															
α_q		0.39	0.22	0.24	-0.06	0.06	-0.02	0.00	-0.12	-0.21	-0.17	-0.04	0.07	0.12	0.26	0.06	-0.04	0.19	0.10	-0.20	-0.16										
β_{MKT}		1.08	1.02	0.96	0.95	0.93	0.91	0.98	0.93	1.04	1.11	1.11	1.05	0.96	1.00	0.98	0.95	0.97	0.93	1.03	1.10										
β_{RM}		0.02	-0.09	-0.12	-0.03	-0.09	-0.04	-0.09	0.05	0.03	0.30	0.23	-0.04	-0.06	-0.10	-0.07	-0.04	-0.12	-0.03	0.09	0.37										
$\beta_{I/A}$		-0.56	-0.09	0.06	0.27	0.22	0.22	0.01	0.01	-0.03	-0.58	0.35	0.26	0.14	0.05	-0.05	0.06	-0.25	-0.30	-0.11	-0.55										
β_{ROE}		-0.26	-0.12	-0.05	0.16	0.00	0.17	0.14	0.11	-0.22	0.03	-0.12	-0.20	-0.04	-0.10	-0.02	0.17	0.09	0.07	0.22	-0.06										
t_q		3.13	1.62	1.02	-0.78	0.66	-0.19	0.00	-1.23	-2.23	-1.86	-0.40	0.75	1.41	2.89	0.71	-0.55	2.54	0.95	-1.91	-1.34										
t_{RM}		32.16	35.50	56.68	48.78	38.87	42.85	44.43	41.06	35.80	39.68	43.24	37.37	30.61	43.73	47.87	40.75	41.32	35.95	50.05	34.83										
$t_{I/A}$		0.46	-2.09	-4.28	-1.17	-2.16	-1.19	-2.55	1.10	0.82	6.79	6.54	-0.91	-1.34	-3.12	-2.22	-1.54	-3.55	-0.85	1.94	7.96										
t_{ROE}		-6.04	-0.80	1.05	5.83	2.75	3.14	0.12	0.08	-0.42	-9.86	6.29	3.76	2.36	0.95	-0.67	1.13	-4.84	-5.17	-1.25	-6.41										
β_{ROE}		-3.55	-1.79	-1.08	4.17	0.05	4.06	3.21	2.13	3.83	0.69	-2.42	-2.70	-0.70	-4.44	-0.46	4.34	2.02	1.23	3.57	-1.24										
Characteristics in the q -factor model																															
ME		0.81	1.48	1.93	2.21	2.37	2.14	2.37	1.78	1.46	0.58	0.90	1.10	1.41	1.69	1.73	1.97	2.47	2.35	1.49	0.54										
I/A		13.87	9.06	7.72	7.37	7.45	7.78	8.63	7.13	11.40	24.02	5.16	6.97	6.48	8.85	8.52	8.38	10.81	11.41	14.22	16.28										
ROE		1.88	2.61	2.90	3.16	3.80	3.09	2.94	2.89	3.75	1.22	1.82	2.24	2.47	2.69	2.21	3.75	3.96	3.51	2.24											

m , a , α_{FF} , and α_C are the average excess returns, the CAPM alphas, the Fama-French alphas, the Carhart alphas, and the α -alphas, and t_m , t_a , t_{FF} , t_C , and t_q their t -statistics, respectively. β_{MKT} , β_{RM} , $\beta_{I/A}$, and β_{ROE} are the average excess returns on the market, the risk-free rate, the I/A ratio, and the ROE, respectively. β_{ROE} is the sum of the ROE and $\beta_{I/A}$.

Panel A: CAPM capitalization (in millions of dollars) of all of the firms in a given decile. Panel B: ROE (in percent) is the sum of the most recently announced quarterly earnings across all of the firms in a given decile divided by the sum of their 1-year-lagged book value. ROE (in percent) is the sum of the most recently announced quarterly earnings across all of the firms in a given decile divided by the sum of their 1-year-lagged book value. ROE is adjusted for heteroskedasticity and autocorrelation.

5.7 Table 10 and Table 11: size-B/M portfolios and Sharpe ratios

Read:

- Table 10 evaluates 25 size and book-to-market portfolios, the classic Fama–French testing assets.
- The q -factor model is competitive with or better than traditional factor models on these portfolios.
- Table 11 shows that the maximum Sharpe ratio for the q -factor model is higher than for the Fama–French and Carhart models.
- The anomaly portfolio set still has a much higher Sharpe ratio, showing that no four-factor model fully spans the anomaly space.

Economic message: The q -factor model offers a stronger investment opportunity set than traditional models, but the anomaly zoo still contains information beyond the four factors.

Table 10
Twenty-five size and book-to-market portfolios

	Low	2	3	4	High	H-L	Low	2	3	4	High	H-L	Low	2	3	4	High	H-L
	m						a (α)=0.29						afp ($\alpha f p$)=0.10					
	bv						t (p)=0.00						tfp (tfp)=0.00					
Small	0.08	0.72	0.84	0.95	1.11	1.02	-0.60	0.13	0.30	0.45	0.59	1.19	-0.54	0.02	0.13	0.18	0.16	0.70
2	0.32	0.69	0.86	0.87	0.99	0.67	-0.34	0.12	0.33	0.39	0.48	0.82	-0.21	0.00	0.09	0.07	0.04	0.25
3	0.38	0.71	0.77	0.77	1.02	0.65	-0.25	0.16	0.28	0.32	0.56	0.81	-0.09	0.04	0.03	0.00	0.13	0.22
4	0.52	0.59	0.73	0.74	0.84	0.32	-0.07	0.08	0.24	0.29	0.40	0.47	0.14	-0.02	0.03	0.00	0.02	-0.12
Big	0.40	0.54	0.54	0.61	0.56	0.16	-0.07	0.10	0.14	0.25	0.19	0.26	0.16	0.11	0.09	0.00	-0.16	-0.32
Small	0.20	2.02	2.57	3.07	3.30	4.59	-2.55	0.64	1.06	2.54	2.98	5.62	-4.84	0.23	1.58	2.53	1.97	5.66
2	0.90	2.20	3.03	3.29	3.31	2.93	-1.94	0.77	2.34	2.77	2.60	3.70	-2.59	0.01	1.36	0.92	0.45	2.19
3	1.15	2.49	3.03	3.05	3.93	2.76	-1.77	1.40	2.27	2.52	2.37	3.63	-1.21	0.52	0.55	0.02	1.28	1.70
4	1.73	2.35	2.91	3.08	3.24	1.43	-0.57	0.76	2.11	2.33	2.53	2.09	1.74	-0.23	0.27	0.03	0.17	-0.87
Big	1.70	2.53	2.71	3.04	2.43	0.79	-0.79	1.27	1.62	1.10	1.17	1.24	2.66	1.29	0.92	-0.02	-1.31	-2.32
	ac (αc)=0.11						afp ($\alpha f p$)=0.11						ME					
Small	-0.48	0.03	0.12	0.18	0.22	0.70	-0.25	0.27	0.31	0.30	0.32	0.57	1.11	0.96	0.92	0.88	0.96	-0.15
2	-0.18	0.03	0.09	0.10	0.04	0.22	-0.14	0.02	0.03	0.07	0.10	0.24	1.14	1.02	1.01	0.94	1.01	-0.13
3	-0.04	0.04	0.09	0.03	0.16	0.20	-0.01	-0.03	-0.04	-0.01	0.14	0.15	1.00	1.05	1.01	0.95	1.01	-0.09
4	0.15	-0.01	0.07	0.03	0.09	-0.06	0.18	-0.14	-0.01	0.02	0.06	-0.12	1.09	1.07	1.06	0.98	1.02	-0.08
Big	0.17	0.07	0.07	-0.03	-0.13	-0.51	0.10	-0.04	0.06	-0.01	-0.04	-0.13	0.98	0.93	0.86	0.90	-0.09	-0.09
	afp ($\alpha f p$)=0.00						afp ($\alpha f p$)=0.00						afp ($\alpha f p$)=0.00					
Small	-2.00	0.36	1.58	2.59	2.53	5.72	-1.48	2.24	3.09	3.68	2.72	2.91	25.50	29.26	36.30	39.11	27.04	-2.63
2	-1.28	0.37	1.34	1.40	0.44	1.88	-1.21	0.29	0.37	0.67	0.89	1.25	33.73	51.06	53.11	48.18	34.08	-2.43
3	-0.50	0.53	0.93	0.28	1.40	1.43	-0.09	-0.30	-0.37	-0.05	1.16	0.92	39.22	53.02	28.69	37.92	28.87	-1.60
4	1.87	-0.16	0.74	0.24	0.75	-0.42	1.50	-1.58	-0.06	0.21	0.44	-0.61	34.34	42.24	31.78	31.00	29.98	-1.30
Big	2.89	0.91	0.70	-0.56	-1.02	-2.12	1.52	-0.49	0.65	-0.06	-0.23	-0.70	52.33	40.01	30.63	30.86	25.72	-2.09

Table 10, part 1

Table 11
Maximum Sharpe ratios

Panel A: Sharpe ratios								Panel B: Maximum Sharpe ratios			
MKT	SMB	HML	UMD	r_{ME}	$r_{I/A}$	r_{ROE}		CAPM	FF	Carhart	q
0.10	0.06	0.13	0.16	0.10	0.24	0.22		0.10	0.21	0.30	0.43
Panel C: Anomaly portfolios											
SUE-1	SUE-6	Abr-1	Abr-6	RE-1	RE-6	R6-6	R11-1	I-Mom	B/M	E/P	CF/P
0.14	0.09	0.23	0.15	0.16	0.13	0.15	0.16	0.10	0.14	0.12	0.11
0.27	0.26	0.28	0.23	0.25	0.21	0.31	0.28	0.20	0.21	0.24	0.21
NSI	CEI	IvG	IvC	OA	POA	PTA	ROE	ROA	GP/A	NEI	FP
0.21	0.14	0.13	0.14	0.10	0.15	0.12	0.15	0.13	0.10	0.14	0.10
0.30	0.32	0.23	0.25	0.25	0.23	0.23	0.24	0.21	0.18	0.26	0.28
NO/P	Dur	I/A	NOA	Δ PI/A	IG						
0.16	0.12	0.11	0.12	0.17	0.14						
0.30	0.24	0.24	0.29	0.25	0.26						
OC/A	Ad/M	RD/M	OL	Svol	All						
0.18	0.14	0.12	0.10	0.14	0.48						
0.26	0.23	0.21	0.17	0.29	1.60						

Panel A reports the monthly Sharpe ratio for each individual factor, calculated as the ratio of the mean factor return to its standard deviation. Panel B reports the maximum monthly Sharpe ratio achievable from each factor model, including the CAPM, the Fama-French (FF) model, the Carhart model, and the q -factor model. The maximum Sharpe ratio is calculated as $\sqrt{\mu_f' V_f^{-1} \mu_f}$, in which μ_f is the vector of mean factor returns for a given factor model, and V_f is the variance-covariance matrix for the vector of factor returns. In Panel C, S_{H-L} is the monthly Sharpe ratio for the high-minus-low decile formed on each anomaly variable, calculated as the mean high-minus-low return divided by its standard deviation. S_m is the maximum monthly Sharpe ratio achievable from a given set of deciles, calculated as $\sqrt{\mu' V^{-1} \mu}$, in which μ is the vector of average returns in excess of the 1-month Treasury bill rate for a given set of deciles, and V is the variance-covariance matrix for the vector of decile excess returns. In column "All," S_{H-L} is the maximum Sharpe ratio achievable from pooling together 28 out of the 35 high-minus-low deciles that have valid data from January 1972 to December 2012, and it is calculated as $\sqrt{\mu_f' V_f^{-1} \mu_f}$, with each high-minus-low decile treated as a different portfolio. S_m is the maximum Sharpe ratio achievable from pooling together 28 out of the 35 sets of deciles that have valid data from January 1972 to December 2012. Table 2 provides a brief description of the symbols. Appendix A details variable definition and portfolio construction.

Table 10
Continued

	Low	2	3	4	High	H-L	Low	2	3	4	High	H-L	Low	2	3	4	High	H-L
	P_{ME}						$P_{I/A}$						P_{ROE}					
Small	1.14	1.18	1.08	1.01	0.99	-0.15	-0.66	-0.32	-0.11	0.17	0.53	1.19	-0.40	-0.39	-0.31	-0.26	-0.35	0.05
2	0.93	0.84	0.83	0.72	0.85	-0.07	-0.72	-0.13	0.25	0.42	0.59	1.11	-0.05	-0.08	-0.02	-0.09	-0.20	-0.14
3	0.72	0.63	0.49	0.43	0.47	-0.24	-0.77	0.03	0.37	0.54	0.78	1.56	0.00	0.05	0.04	-0.09	-0.16	-0.16
4	0.40	0.33	0.24	0.20	0.18	-0.21	-0.70	0.18	0.39	0.60	0.77	1.47	0.04	0.09	-0.01	-0.14	-0.16	-0.21
Big	-0.22	-0.09	-0.21	-0.10	-0.14	0.08	-0.39	0.12	0.30	0.61	0.80	1.20	0.14	0.16	-0.05	-0.06	-0.25	-0.39
	$P_{I/A}$						$P_{I/A}$						P_{ROE}					
Small	16.49	18.22	23.39	33.68	14.79	-1.31	-5.09	-3.64	-1.46	3.06	5.90	8.39	-3.34	-4.73	-5.43	-7.53	-6.82	0.40
2	16.68	32.58	19.26	12.82	13.78	-0.70	-9.48	-2.32	3.93	6.26	8.97	10.46	-0.68	-1.84	-0.56	-1.76	-3.24	-1.25
3	14.60	16.16	5.21	7.05	4.85	-1.80	-11.40	0.43	3.35	6.24	9.13	12.41	0.02	0.93	0.54	-1.23	-2.08	-1.50
4	6.26	6.97	3.22	3.63	2.14	-1.54	-8.47	2.51	4.05	5.30	7.28	9.57	0.68	1.62	-0.19	-1.82	-1.98	-1.65
Big	-7.59	-2.87	-5.22	-1.63	-2.15	1.06	-9.47	2.19	4.98	4.87	6.35	8.00	4.37	3.89	-0.89	-0.68	-2.93	-3.99
	ME						I/A						ROE					
Small	0.07	0.07	0.07	0.06	0.05	-0.03	10.90	12.67	9.00	7.02	1.77	-9.13	-1.38	0.65	0.79	0.56	-0.91	0.46
2	0.33	0.33	0.33	0.33	0.32	-0.01	17.87	14.63	11.22	7.57	3.92	-13.94	2.10	2.34	2.01	1.47	0.05	-2.05
3	0.76	0.76	0.76	0.76	0.76	0.00	18.30	13.64	9.95	8.14	4.47	-13.82	3.35	2.86	2.20	1.63	0.65	-2.70
4	1.86	1.83	1.78	1.77	1.82	-0.04	15.28	11.31	8.86	7.06	5.18	-10.09	4.24	3.14	2.28	1.58	0.74	-3.50
Big	15.98	13.55	11.19	9.94	8.45	-7.53	12.59	11.36	7.49	8.22	6.44	-6.14	5.44	3.71	2.64	2.24	1.05	-4.38

$m, a, afp, ac,$ and afp are the average excess return, the CAPM alpha, the Fama-French alpha, the Carhart alpha, and the q -alpha, and $t_m, t_{afp}, t_c,$ and t_p are their t -statistics, respectively. afp ($\alpha f p$), ac (αc), and afp ($\alpha f p$) are the average magnitude of the alphas, and $p_{afp}, p_{ac},$ and p_{afp} are the p -values of the GRS test across the 25 portfolios. $P_{ME}, P_{I/A}, P_{ROE}, P_{I/A},$ and P_{ROE} are the market size, momentum, and ROE factor loadings in the afp -factor model, and $t_m, t_{afp}, t_c,$ and t_p are their t -statistics, respectively. ME is the market return contribution.

Table 10, part 2

6 Short summary

- The paper proposes a four-factor investment-based asset-pricing model.
- The factors are market, size, investment, and profitability.
- Investment-based theory predicts lower expected returns for high investment firms and higher expected returns for high profitability firms.
- The paper tests nearly 80 anomalies across six categories.
- Many anomalies are insignificant in broad NYSE-breakpoint value-weighted deciles.
- Among significant anomalies, the q -factor model absorbs many alphas that remain under CAPM, Fama–French, and Carhart models.
- The model works because anomaly portfolios load on investment and ROE factors.

- The model has the most difficulty with short-horizon momentum and some accrual specifications.
- It performs well on classic size/book-to-market portfolios and has a high maximum Sharpe ratio.
- The main lesson is that investment and profitability provide a compact economic structure for digesting a large anomaly zoo.

7 Conclusion

7.1 Core conclusion

Hou, Xue, and Zhang argue that many return anomalies can be organized by a four-factor q -factor model grounded in investment-based asset pricing. The model's investment and profitability factors absorb a large fraction of anomaly returns that traditional models leave unexplained.

7.2 Economic intuition

The intuition is simple: expected returns are tied to firms' investment and profitability decisions. Firms that invest aggressively tend to have lower expected returns, while firms with high profitability tend to have higher expected returns. Many anomaly variables are correlated with these two forces.

7.3 Takeaway

The paper does not eliminate every anomaly, but it substantially reduces the dimensionality of the anomaly zoo. It shifts the asset-pricing conversation from isolated characteristics toward a small set of economically interpretable factors.